# **Denying the Play**: Frame-level Analysis of Passing Lanes and Defensive Pressure in Hockey

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## 1 Introduction

Ice hockey is a dynamic, continuous game where the success of a play depends on the interaction between passing decisions, player positioning, and defensive pressure. While passes are central to offensive creation, their effectiveness is shaped not only by the puck carrier but also by receiver positioning and the pressure applied by defenders. Traditional event-based metrics focus on outcomes such as completed passes or turnovers, but they do not capture the spatial context in which these actions occur. As a result, they overlook how defenders constrain passing options, how receivers create opportunities, and how pressure influences decisions in real time. Recent work using tracking data has begun to address these gaps. Radke et al. [2] modeled passing opportunities through passing lanes, while Andrienko et al. [1] quantified defensive pressure using spatial proximity and movement.

Building on these ideas, we propose a frame-level framework to evaluate pass quality, pressure on puck carriers and receivers, receiver ability, and defender performance. By integrating these components, our approach provides a unified and actionable view of how teams create and restrict opportunities in ice hockey.

In this work, we address the following questions:

- How does pass quality vary as a function of pressure on the puck carrier and intended receiver?
- How does receiver positioning and movement influence the creation of high-quality passing opportunities?
- How effectively do defenders reduce pass quality by obstructing passing lanes and applying pressure?
- How do teams differ in their ability to generate high-quality passing opportunities under defensive pressure?

## 2 Methodology

### 2.1 Passing Lanes

A passer can utilize four distinct passing lanes to deliver the puck to teammates, assuming full strength, excluding the goalie. We consider the passing lane framework from [2], which carefully considers the geometry of the puck carrier (p), receivers (r), and opponents (o) to construct a passing lane (Fig.1). Each passing lane is determined by a non-negative real numbered parameter **γ**, which corresponds to the size and openness of the passing lane. Fig. 2 shows passing lane examples from the plays in the data. Access the app for play animations in the Appendix **[Link]**.

***Fig. 1*** *Passing Lanes Framework. p is the puck carrier, r is the potential receiver, and o are the opponents. The value of* ***γ*** *determines the size/open-ness of the passing lane. See [2] for more details.*

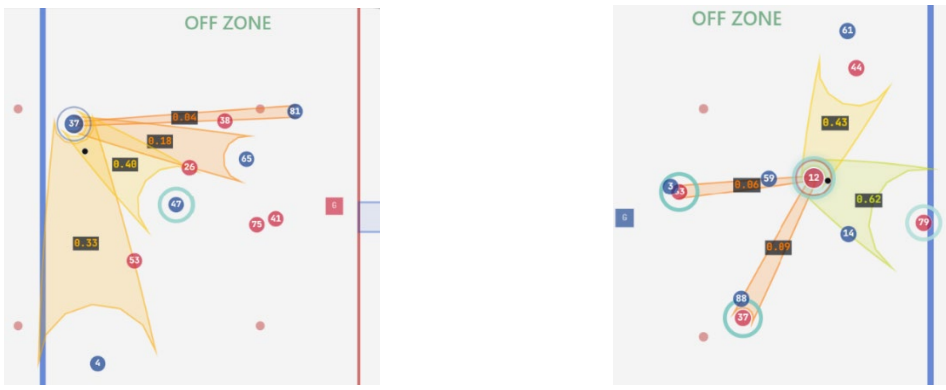



***Fig. 2*** *Passing Lanes Examples from the data. We didn't add circles around players for clarity. The numbers represent* $\gamma$ *(size/open-ness of the passing lane). The ring and color intensity around receivers represent the pressure exerted by the nearby defenders on that receiver. These play animations can be accessed through the Pressure Shadow APP* ***[Link]****, see Appendix for details.*

## 2.2 Pressure

Exerting pressure on a puck carrier (p) is a strategic tactic employed by one or more opponents (o) to encourage mistakes from the player in possession. Additionally, potential receivers (r) can also face pressure from nearby opponents, impacting their decision-making. Analyzing this pressure can provide valuable insights into how players respond to various game scenarios, how they position themselves, and whether they are reacting swiftly enough to changes on the ice.

We use a modified version, as in [2], of Zone of Pressure (ZoP) defined for Soccer by [1]. We consider a smaller ZoP since playing surface and team formations are smaller in ice hockey. The pressure boundary is calculated the same way as in [1], resulting in an oval structure. The limits of ZoP behind ($d_{back}$ = 5ft) and in front ($d_{front} = 3*d_{back}$) of p are adjusted for ice hockey. We extend the notion of $V^{threat}$ from [1] and [2], see Appendix for details. The total pressure experienced by p with o pressers is evaluated as $Pr(p) = \sum_o o(p)$. Similar calculation is done to evaluate the pressure exerted by o on receivers.

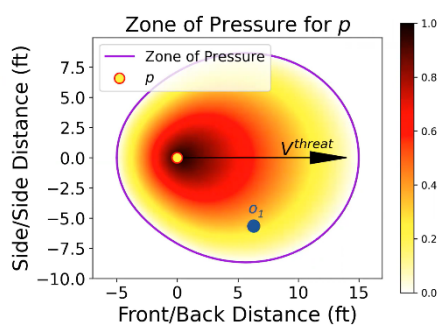



***Fig. 3*** *Zone of pressure (ZoP) around p, direction of* $V^{threat}$*, and pressure p experiences from o* $\epsilon$ *(0, 1) regarding the location of o. Darker corresponds with more pressure.*

## 2.3 Novel Metrics

Fig. 4 provides a glimpse of the passing lane and pressure setup. Using passing lane and pressure framework, we define a range of new metrics for passing effectiveness, receiver and puck carrier ability, defense performance and pressure effect on player movement. See Table 1 for more details.

*OPS — Open Pass Score:* A frame-level metric quantifying how much passing opportunity a puck carrier has at any given moment, for each available receiver. It combines three factors: how open the passing lane is to the available receiver ($\gamma$, the lane quality parameter), how much defensive pressure that receiver is under (ZoP_r), and how much pressure the carrier themselves is under (ZoP_c): $OPS = (1 - e^{-\gamma}) \cdot e^{-ZoP_r} \cdot e^{-\alpha \cdot ZoP_c}$

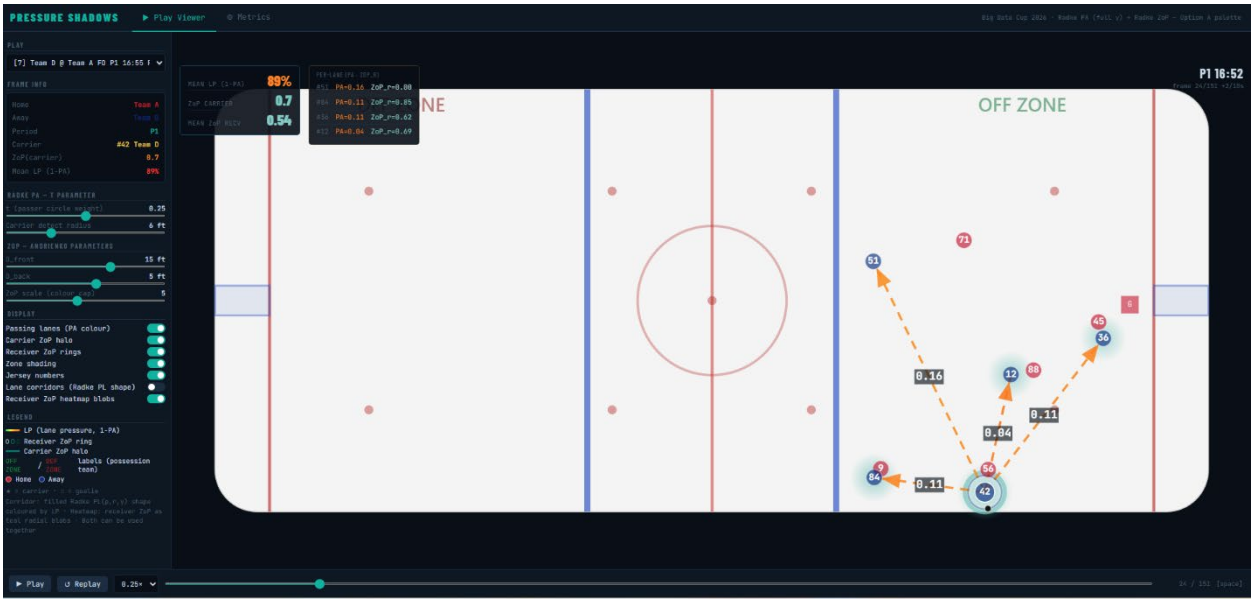



***Fig. 4*** *A snapshot of how passing lanes and pressure come together. A snapshot of the play showing the passing lanes as dotted lines, lane openness by gamma value displayed on lane, pressure on receiver and puck carrier by luminance of teal color. The stats are also shown in the top left corner. These play animations can be accessed through the Pressure Shadow APP* ***[Link]****, see Appendix for details. There are toggle options to display full pass lanes as shown in Fig. 2 along with other viewing settings.*

Values range from 0 to 1. A value near 1 means a wide-open lane to an unmarked receiver who is also under no pressure from the carrier's side. A value near 0 means either the lane is blocked, the receiver is heavily marked, or the carrier is under heavy pressure. α controls how much carrier pressure discounts the opportunity. Setting α=0.5 means carrier pressure is half as penalizing — reflecting the fact that an experienced carrier can still make a clean pass under moderate pressure, while a receiver who is marked cannot easily get open. We use α=0.5, however with more data we can validate and select a different value. OPS is computed at 10fps for every carrier frame in 5v5 play. Access OPS Play App**[Link]** for play animations, see Appendix for more details.

*Receiver OPS:* Similar to OPS metric with the carrier pressure term removed: Receiver OPS = $(1 - e^{-\gamma}) \cdot e^{-ZoP_r}$. This isolates the receiver's contribution to passing opportunity — how open the lane is to them and how free they are — independent of what is happening to the carrier.

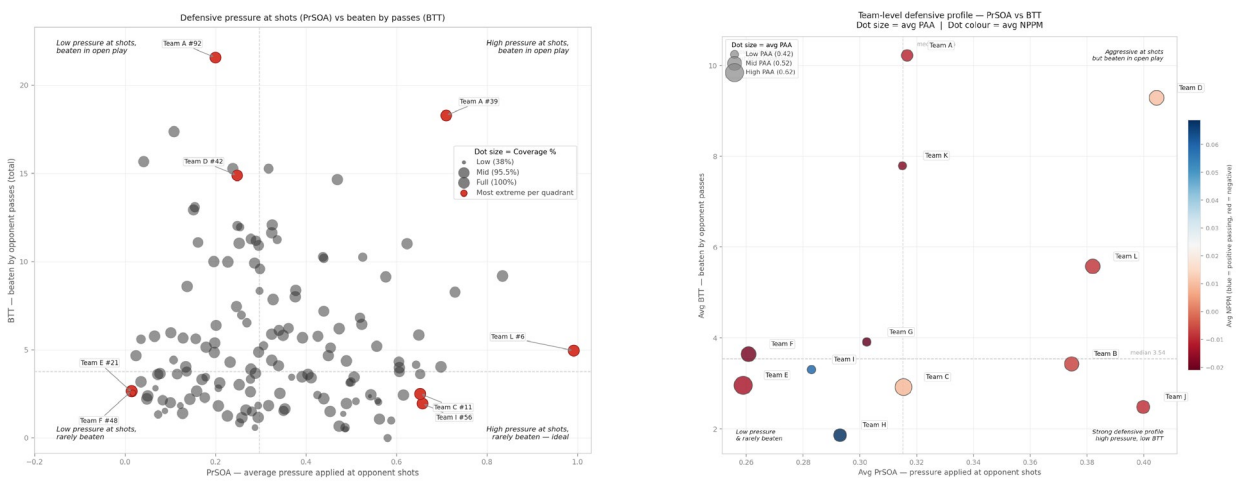
| **Name (Sym)** | **Description** |
|---|---|
| Passing Availability Average (PAA) | Measures the average size of the passing lane (γ value) for a player's completed passes. |
| Pressure at Shots Against (PrSOA) | The average pressure a player applies to an opposing shooter at the moment of a shot attempt. |
| Beaten Total (BTT) | How often have opponents' passes bypassed this player. |
| Overtaken Total (OVT) | The cumulative fraction of opponents a player's passes have advanced the puck beyond. |
| Normalized Passing Plus-Minus (NPPM) | The normalized version of PPM (OVT minus BTT), divided by the number of passing events that contributed. |
| Open Pass Score (OPS) | Measures the quality of a pass option considering passing lane quality, receiver and carrier pressure |
| Receiver Open Pass Score (Recv OPS) | Measures how open the receiver is |
| Open Availability Rate (OAR) | fraction of off-puck frames where Receiver OPS > 0.25 |
| Player Coverage Rate (PCR) | Evaluates how well a defensive player restricts passing lanes and applies pressure that forces opponents into difficult or suboptimal decisions. See Appendix for more detail. |
| Defensive Chaos (DC) | Measures how often a defensive player forces a poor play. This identifies players who capitalize on pressure by triggering turnovers or takeaways. See Appendix for more detail. |

*Table 1: Name and description of the metrics.*

# 3 Findings

We present our findings in this section based on the passing lane and pressure framework using the new defined metrics. Fig. 5a compares PrSOA and BTT to evaluate each player's defensive profile across four quadrants defined by the dataset medians, with dot size reflecting defensive coverage. The ideal profile — high shot pressure with few passes conceded — sits in the bottom-right, while the top-left represents the greatest liability. Team I #56 (PrSOA=0.657, BTT=1.950) exemplifies disciplined defense, closing hard on shots while rarely being bypassed, proving to be a sound defender. Team L #6 (PrSOA=0.991, BTT=4.950) is the most aggressive shot-presser in the dataset, but concedes passes at a moderate rate, suggesting their forward aggression allows opponents to exploit the space left behind.

Fig. 5b aggregates these profiles to the team level, plotting average PrSOA against average BTT with dot size encoding average PAA and dot color encoding average NPPM. Team D and Team J occupy the strong defensive bottom-right quadrant with above-median shot pressure and below-median BTT, though their negative NPPM suggests a defensive-first system that prioritizes shot pressure over offensive passing output. Team H is the standout, combining the highest average NPPM with below-median BTT — winning the passing battle while maintaining defensive discipline. Team A and Team K sit in the most vulnerable quadrant, consistent with the liability profiles identified in Figure 5a. Together, Figures 5a and 5b demonstrate that PrSOA and BTT reveal defensive variation at both the individual and team level that traditional statistics would not capture.




**(a)** **(b)**

***Fig. 5*** *A comparison of PrSOA and BTT at the (a) individual player level, where dot size represents the coverage percentage of p and dots highlighted in red denote the most extreme player within each quadrant, and (b) team level, where dot size represents average PAA and dot color represents average NPPM, revealing how individual defensive profiles compound into systemic team-level defensive tendencies.*

Fig. 6a cross-references PrMA against TOA to examine whether players who carry the puck under high defensive pressure are also more prone to turnovers. Blue circles denote positive NPPM players, and green squares denote negative NPPM players, with median lines dividing the plot into four quadrants. Most of the dataset clusters below the PrMA median with low-to-moderate turnovers, suggesting most players carry under moderate pressure without frequently conceding possessions. Team F #29 sits in the high-TOA, low-PrMA quadrant with 26.667 turnovers despite carrying under light pressure, pointing to a decision-making concern rather than a pressure-induced one. Fig. 6b replaces PrMA with PAA to examine whether pass lane quality relates to turnover rate. Team F #29 again emerges as an outlier, combining a high PAA of 0.781 with a high TOA of 26.667 — finding consistently open

passing lanes yet still conceding possession frequently, suggesting pass volume rather than poor lane selection drives their turnover rate. Together, Figures 3a and 3b mirror the cross-referencing approach of [2], demonstrating that PrMA, PAA, and TOA in combination reveal passing behaviors that neither metric capture independently.

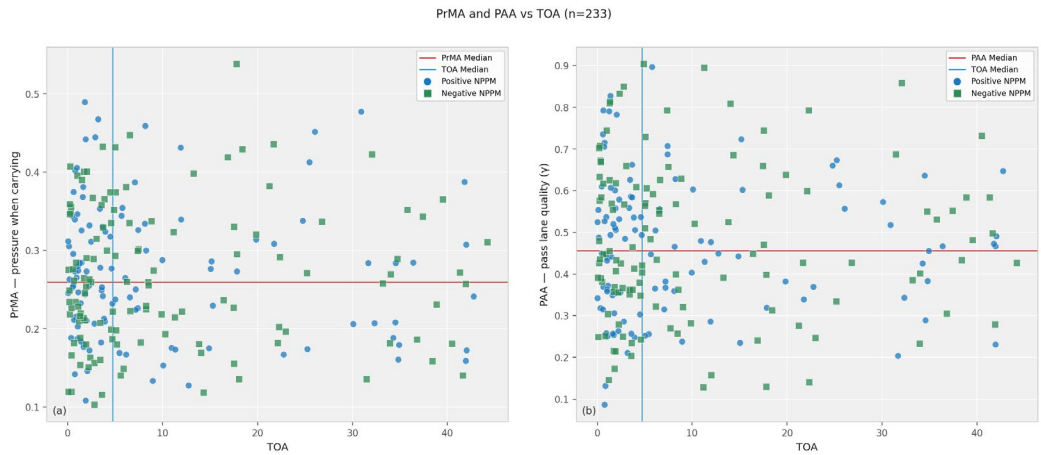
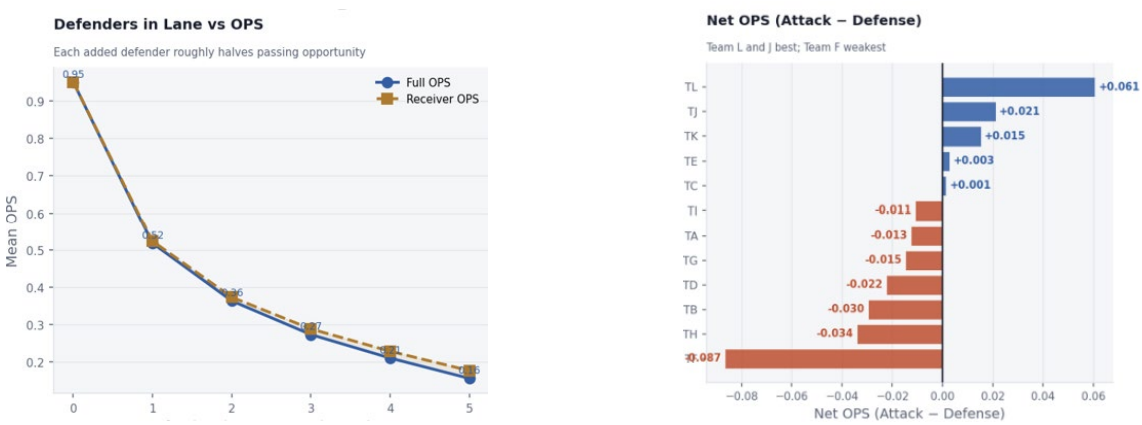



***Fig. 6*** *Cross-referencing PrMA and PAA against TOA, where blue circles denote positive NPPM players and green squares denote negative NPPM players*

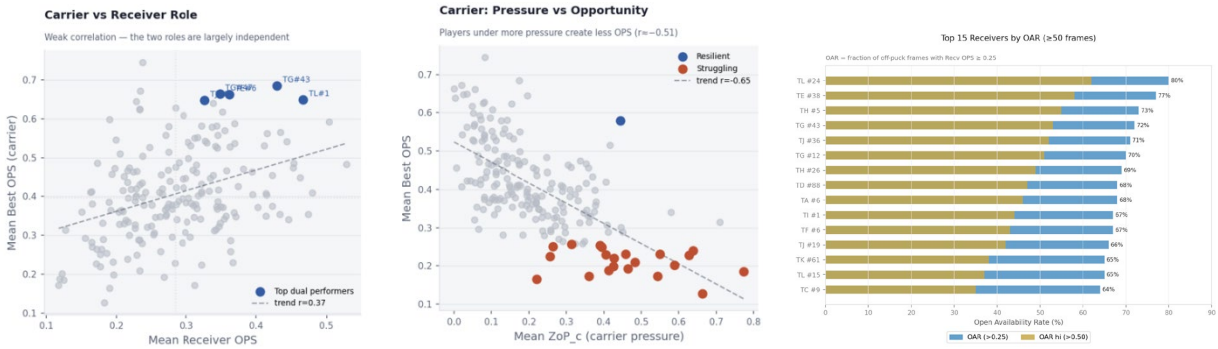



*(a)* *(b)*

***Fig. 7*** *Effect of number of defenders on Open Pass Score (OPS) and the net OPS for teams.*

Line chart, Fig. 7a, shows the effect of defensive lane congestion on passing opportunity. Each additional defender between the carrier and the net roughly halves both Full OPS and Receiver OPS. Diverging bar chart of Net OPS (Attack minus Defense) for each team. Positive values (blue) indicate teams whose offensive passing output exceeds what they concede, while negative values (red) indicate the reverse. Team L and Team J show the strongest and weakest net performance, respectively.

As shown in Fig. 8a, the scatter plot comparing individual player performance as a carrier (Mean Best OPS) vs. as a receiver (Mean Receiver OPS) ($r = 0.37$) reveals that the two roles are largely independent, with blue dots highlighting elite dual performers who rank highly in both dimensions. Fig. 8b further demonstrates that players under greater defensive pressure tend to generate fewer high-quality passing opportunities ($r = –0.51$), with red dots indicating struggling carriers and a single blue dot marking a notably resilient outlier. Finally, Fig. 8c ranks the top 15 receivers by Open Availability Rate (OAR) among players with ≥50 off-puck frames, where OAR represents the fraction of frames where a receiver's OPS exceeded 0.25 (blue) or 0.50 (gold).




*(a)* *(b)* *(c)*

***Fig. 8*** *Carrier and receiver passing performance*

Using DC and PCR, we construct defensive archetypes that characterize how each player contributes defensively. As shown in Figure 9, players are split across four quadrants by median DC and PCR thresholds into three primary profiles: Elite defenders combine high coverage with high chaos to disrupt both carriers and passing lanes; Disruptors generate high chaos with lower coverage, reflecting aggressive but positionally selective play; and Suppressors maintain strong coverage without generating significant chaos, representing a disciplined lane-blocking style. Passive players in the bottom-left contribute minimally across both dimensions. This framework reveals not just how effective a defender is, but what style of defense they employ.

# 4 Discussion

We have extended the frameworks introduced by the [1] and [2] to analyze the player movement and behavior under pressure at frame-level. Here are a few key points of action that are supported by our analysis of pass quality, pressure, and player positioning.

- **Use OPS to review passing decisions at intermissions or postgame**
  Players and coaches can leverage Open Pass Score (OPS) thresholds (e.g., >0.25 or >0.50) to identify when passing opportunities are truly advantageous versus forced.
- **Balance aggression with positional discipline**
  High shot pressure (PrSOA) can be effective, but overly aggressive defenders risk being bypassed (high BTT). Teams should optimize pressure timing to avoid creating exploitable gaps.
- **Exploit defensive archetypes in opponent scouting**
  Classifying defenders as *Elite*, *Disruptors*, or *Suppressors* (via PCR and DC) enables tailored offensive strategies—e.g., quick puck movement vs. disruptors, patience vs. suppressors.
- **Balance aggression with positional discipline**
  High shot pressure (PrSOA) can be effective, but overly aggressive defenders risk being bypassed (high BTT). Teams should optimize pressure timing to avoid creating exploitable gaps.

We have laid out the fundamental framework using passing lanes and pressure to analyze player movement and behavior at the frame-level. There are several ways to extend our project. One significant extension would be to incorporate player's speed and direction of movement to create realistic passing lanes and estimate the pressure suitably.

# References


[1] G. Andrienko, N. Andrienko, G. Budziak, J. Dykes, G. Fuchs, T. V. Landesberger, and Hendrik Weber. *Visual analysis of pressure in football.* Data Mining and Knowledge Discovery, 31:1793–1839, 2017.

[2] Radke, D.T., Radke, D.L., Brecht, T., Pawelczyk, A. *Passing and pressure metrics in ice hockey*. AI for Sports Analytics Workshop (2021)

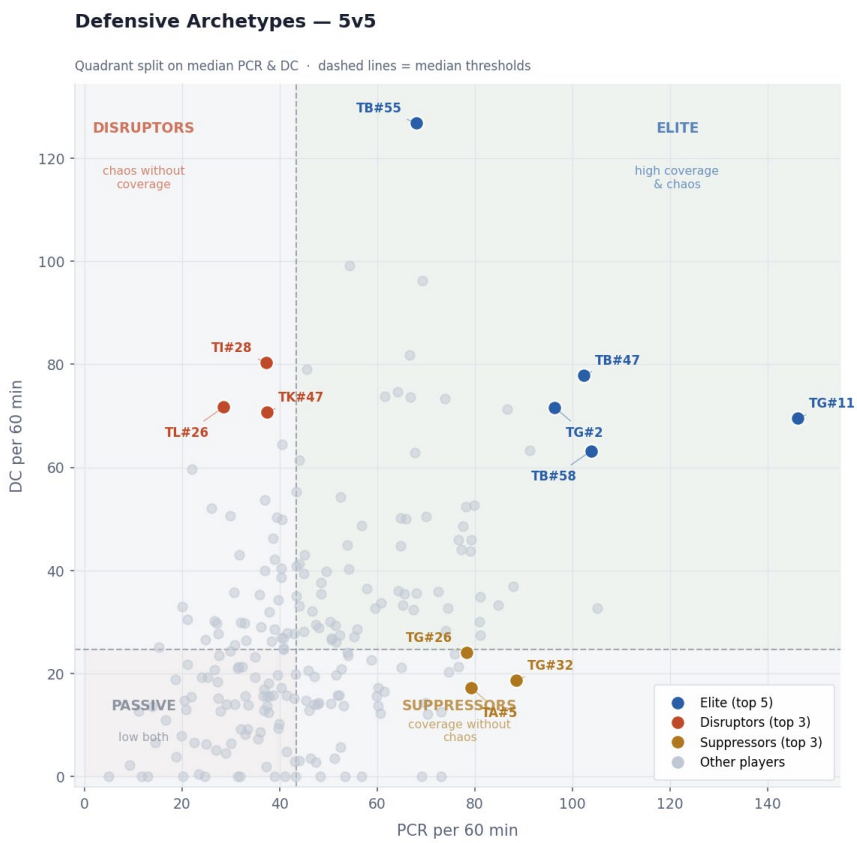




***Fig. 9*** *Defensive Archetypes identified using pressure coverage rate (PCR) and defensive chaos (DC).*

# Appendix

- **CODE Link**
- **Pressure Shadow App [Link]:** Play animations with passing lanes, pressure and other settings. See Fig. 4.
    - Select different plays on top left.
    - Select different settings for the display.
- **OPS Play Animation App [Link]:** Play animations for different scenarios with OPS related metrics such as real time defense pressure tracker, peak opportunity highlight, peak opportunity duration, and receiver availability.
    - Select different plays on the top.
    - Four sub-plots updating in real time:

- **Peak Opportunity** — running time series of best_ops with ZoP_c as a red filled area behind it, peak moment marked with a yellow vertical line and label. Togglable between full series and rolling 6-second window.
- **Opportunity Duration** — cumulative seconds above 0.25 or 0.50 threshold (toggle between them), with vertical ticks at each threshold crossing.
- **Receiver Availability** — one coloured line per receiver, white thicker line for best available OPS. Togglable to mini rink view showing receivers coloured by their current OPS.
- **Defensive Pressure** — gauge showing current ZoP_c with a mini time series, or cumulative heatmap on a mini rink showing where the carrier has faced the most pressure throughout the episode.

***Fig. 10*** *A snapshot of the OPS Play Animation App, with realtime plots of peak opportunity, opportunity duration, receiver availability and defensive pressure gauge.*

- **$V^{threat}$ Adjustment for Pressure:** We extend the notion of threat directions $V^{threat}$ as described in [1] and [2]. The target (puck carrier or receiver) faces multiple directions from which they can advance the puck toward the opponent's goal. We define a set of threat vectors $V^{threat}$ as vectors pointing from the target toward each threatening destination. Rather than just one $V^{threat}$ towards the center of the opening to the attacking net, which is always included as a threatening direction. The additional threat vectors are pointing towards potential receivers who are closer to the puck carrier than the goal. We simplify the formula used for the pressure exerted by o onto p, to scale linearly, so that the pressure o exerts on p inside of the ZoP in direction of $V^{threat}$, $o(p_{V^{threat}}) = 1 - \frac{\text{dist}(p_{V^{threat}},\ \text{o})}{\text{dist}(p_{V^{threat}},\ \text{z})}$, where z is the point on the ZoP boundary in the direction of o. The pressure exerted by o onto p, $o(p) = \max_{V^{threat}} o(p_{V^{threat}})$.

- **PCR — Player Coverage Rate**
  - Measures how much a defender suppresses passing opportunity each time the opponent carries the puck, normalised per 60 minutes of 5v5 ice time. At each tracking frame where a defender is within 60ft of the carrier, their individual contributions to two ZoP values are computed: how much pressure they place on each potential receiver (ZoP_r, weighted 75%) and how much

pressure they place on the carrier (ZoP_c, weighted 25%). These per-frame contributions are summed across all defending frames and divided by 5v5 defending time in minutes, then multiplied by 60.

- **DC — Defensive Chaos**
  - Measures how often a defender's pressure directly precipitates a turnover or forced error, normalised per 60 minutes of 5v5 ice time. A chaos event is triggered whenever the total carrier ZoP_c exceeds 0.5 and an Incomplete Play or Takeaway occurs within 2 seconds. Credit is distributed proportionally among all defenders contributing to that ZoP_c spike — a defender responsible for 60% of the total ZoP_c at the moment of the spike receives 60% of the credit for the resulting chaos event. Credits are summed and divided by 5v5 defending time per 60.

Composite Defense Rating (0–1) for each team, combining OPS allowed, ZoP_c applied, and ZoP_r applied into a single ranked score.